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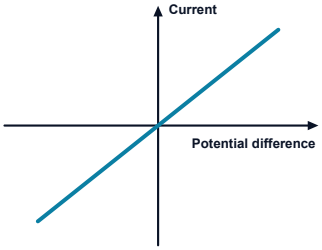
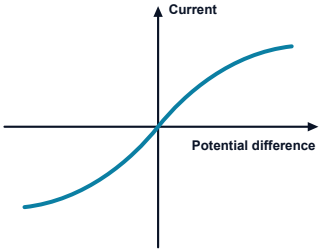
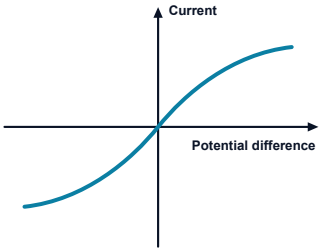
Current, resistance and potential difference

Electricity
Paper 1 · Combined and Separate
24 questions

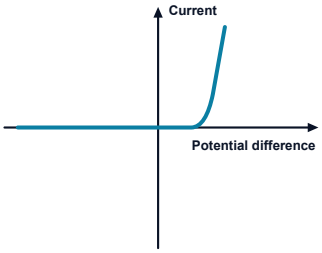
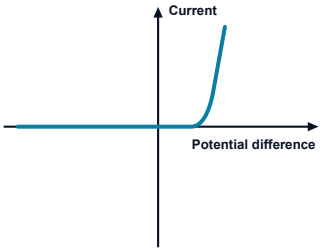
Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 12 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**What is an electrical conductor?**	A material that allows charge to flow through it.	☐
2	**Name the charged subatomic particles that can flow in metals.**	Electrons	☐
3 ★	**Define electric current.**	The rate of flow of electrical charge.	☐
4 ★	**State the standard units of charge, current and time.**	Charge: coulomb (C) Current: ampere (A) Time: second (s)	☐
5 ★	**Describe the current at different points in a single closed loop.**	It is the same at every point.	☐
6	**Name the two variables that the current through a component depends on.**	The potential difference across it and its resistance	☐
7	**The potential difference across a fixed resistor is increased. Describe the effect on the current.**	The current increases.	☐
8	**The resistance of a component is increased at a constant potential difference. Describe the effect on the current.**	The current decreases.	☐
9 ★	**State the standard units of potential difference and resistance.**	Potential difference: volt (V) Resistance: ohm (Ω)	☐
10	**Name the type of component whose resistance stays constant as the current changes.**	An ohmic conductor (at constant temperature)	☐

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	QUESTION	ANSWER	✓
11 ★	**Describe how current depends on potential difference for an ohmic conductor.**	Current is directly proportional to potential difference.	☐
12	**Name the type of component that gives this I–V graph.** 	An ohmic conductor (e.g. a fixed resistor)	☐
13 ★	**Name the component that gives this I–V graph.** 	A filament lamp	☐
14	**This is the I–V graph for a filament lamp. Describe how its resistance changes as the potential difference increases.** 	The resistance increases.	☐
15 ★	**Explain why the resistance of a filament lamp increases with current.**	A greater current makes the filament hotter, and its resistance increases with temperature.	☐

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	QUESTION	ANSWER	✓
16 ★	**Name the component that gives this I–V graph.** 	A diode	☐
17 ★	**This is the I–V graph for a diode. Describe its resistance when the potential difference is reversed.** 	Very high (almost no current flows).	☐
18 ★	**Describe how the resistance of a thermistor varies with temperature.**	As the temperature increases, the resistance decreases.	☐
19 ★	**Describe how the resistance of an LDR varies with light intensity.**	As the light intensity increases, the resistance decreases.	☐
20	**Give one use of a thermistor and one use of an LDR.**	Thermistor: thermostat LDR: lights that switch on when it gets dark	☐
21	**State the two things needed for electrical charge to flow in a circuit.**	A complete circuit and a source of potential difference.	☐
22	**State the equation linking charge flow, current and time.**	charge flow = current × time (Q = I t)	☐
23	**State the equation linking potential difference, current and resistance.**	potential difference = current × resistance (V = I R)	☐
24 ★	**State the condition for the current through a resistor to be directly proportional to the potential difference.**	Its temperature must stay constant.	☐

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